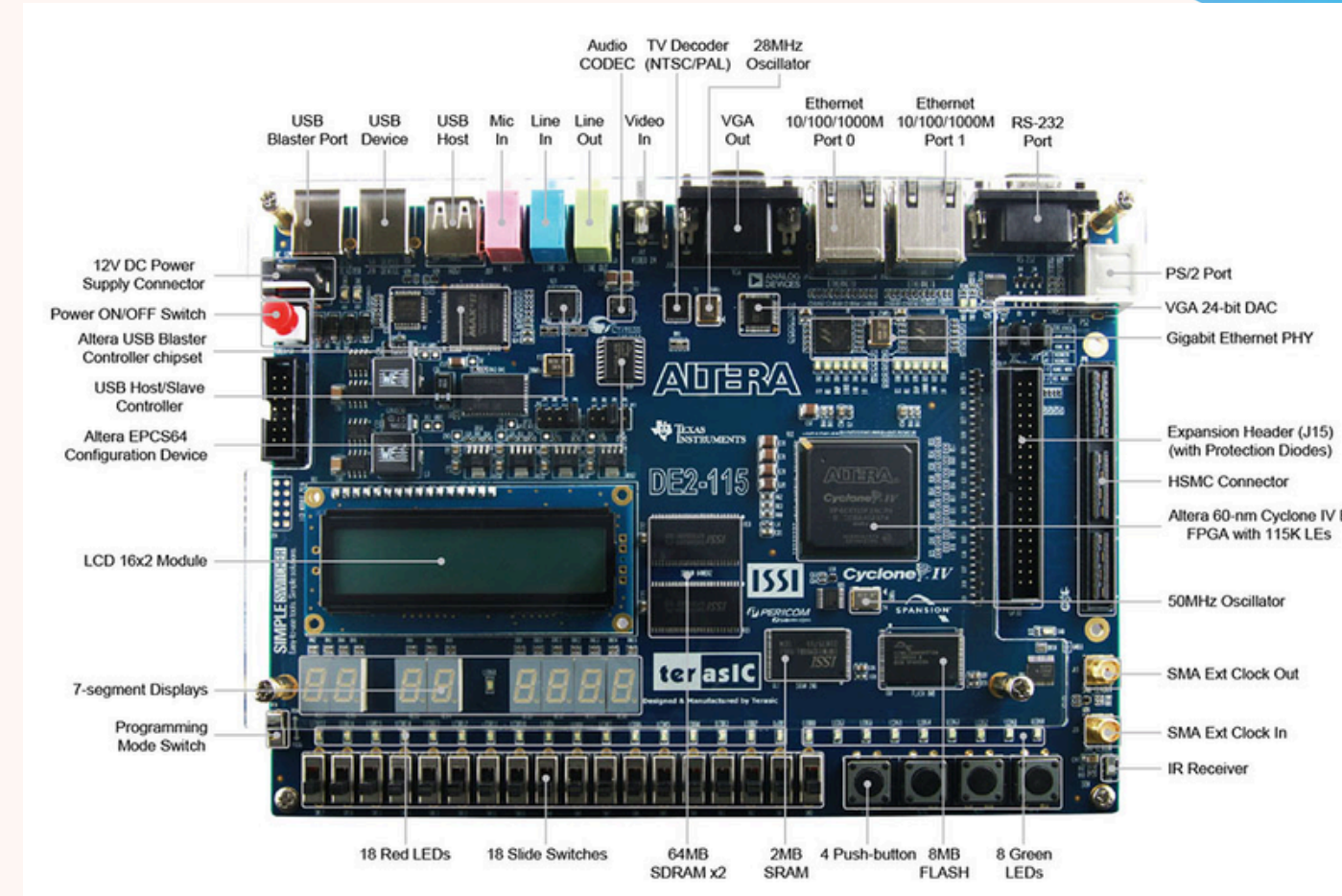
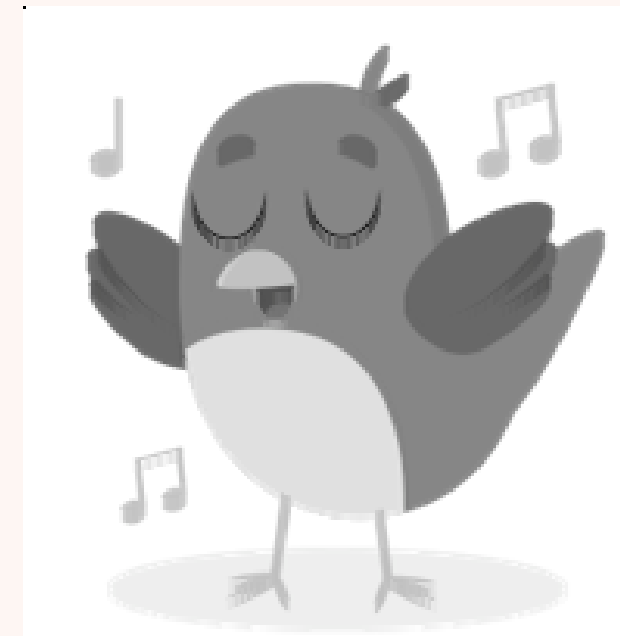
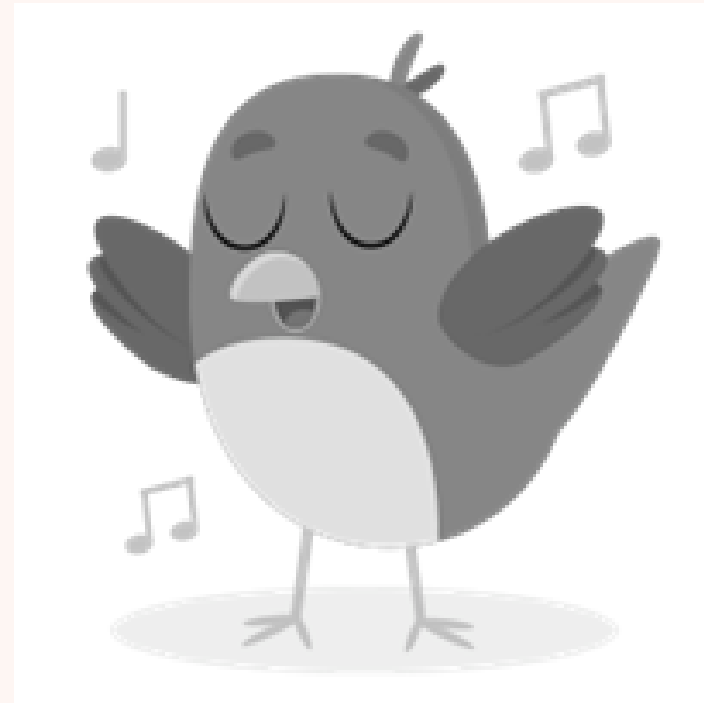




FILTRO MEDIANA EN FPGA

MICRO/NANO SISTEMAS ELECTRONICOS

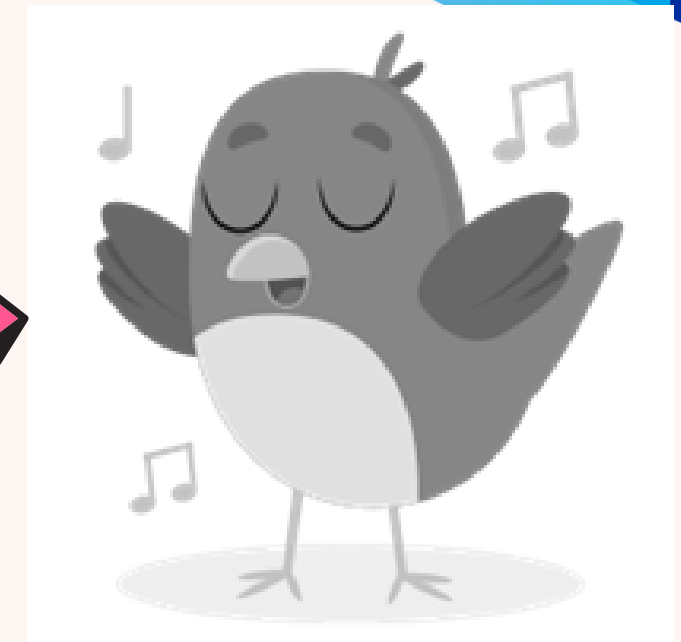
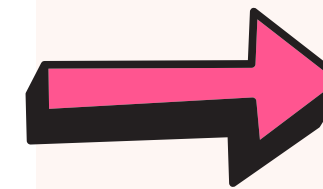
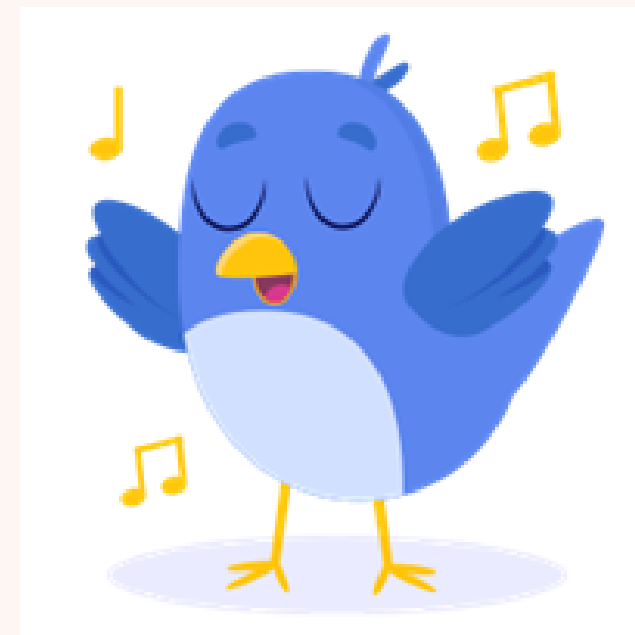
INTRODUCTION



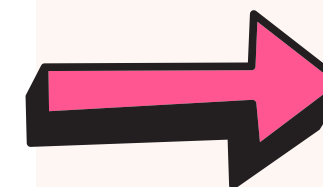
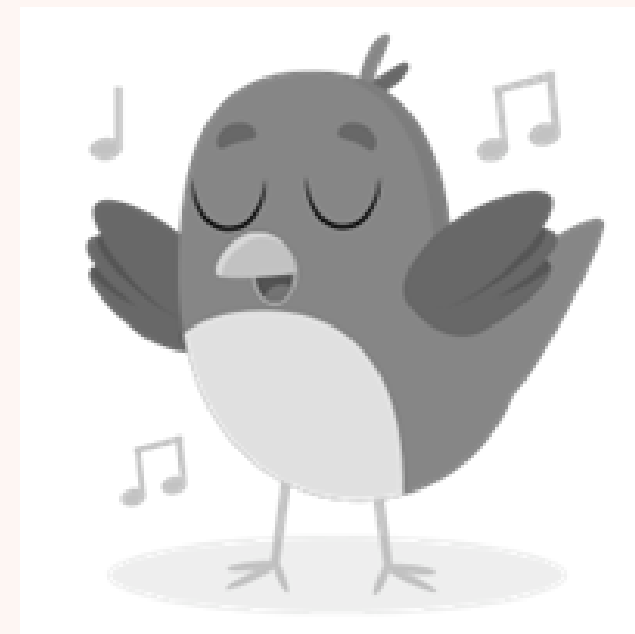
PROCESAMIENTO DE IMAGEN



$I = \text{RGB2GRAY}(I)$

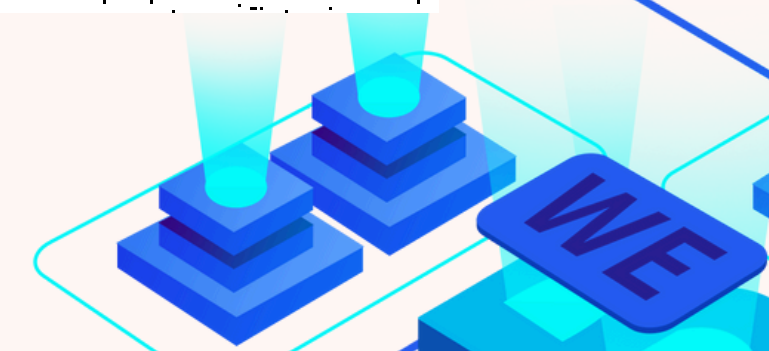
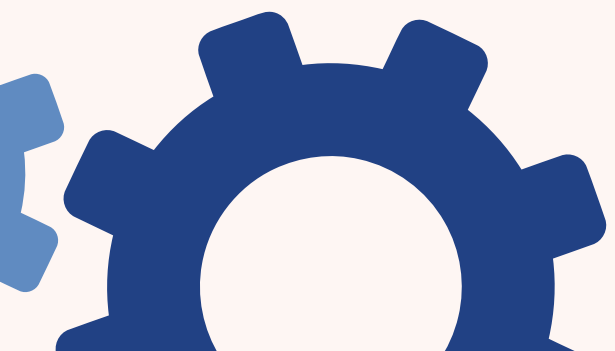


$J = \text{IMNOISE}(I, \text{TIPO}, \text{PARÁMETRO})$

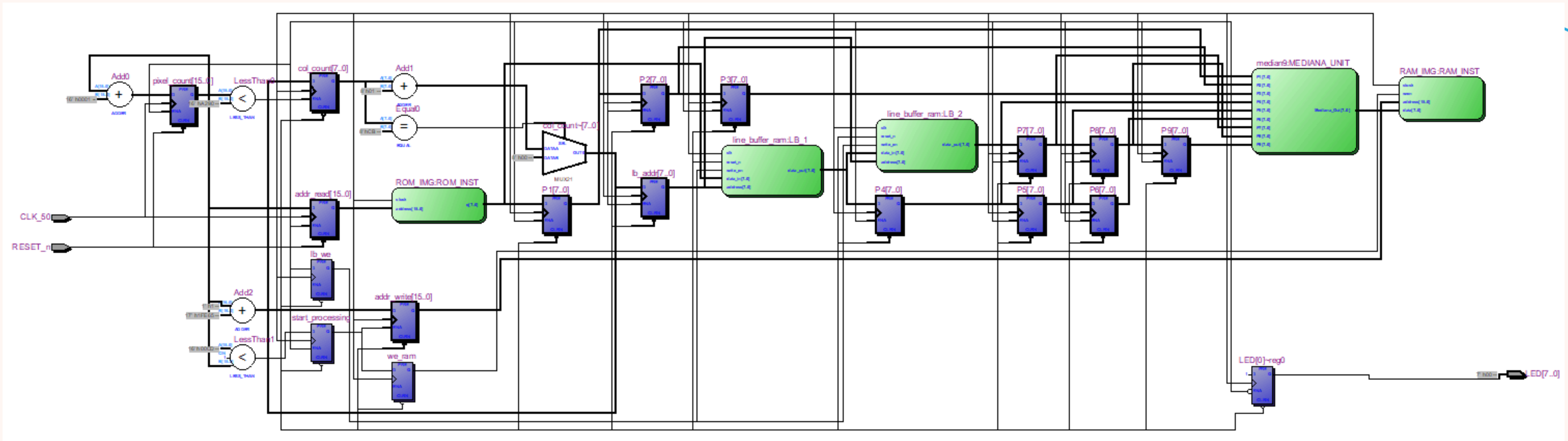


GENERATE_MIF

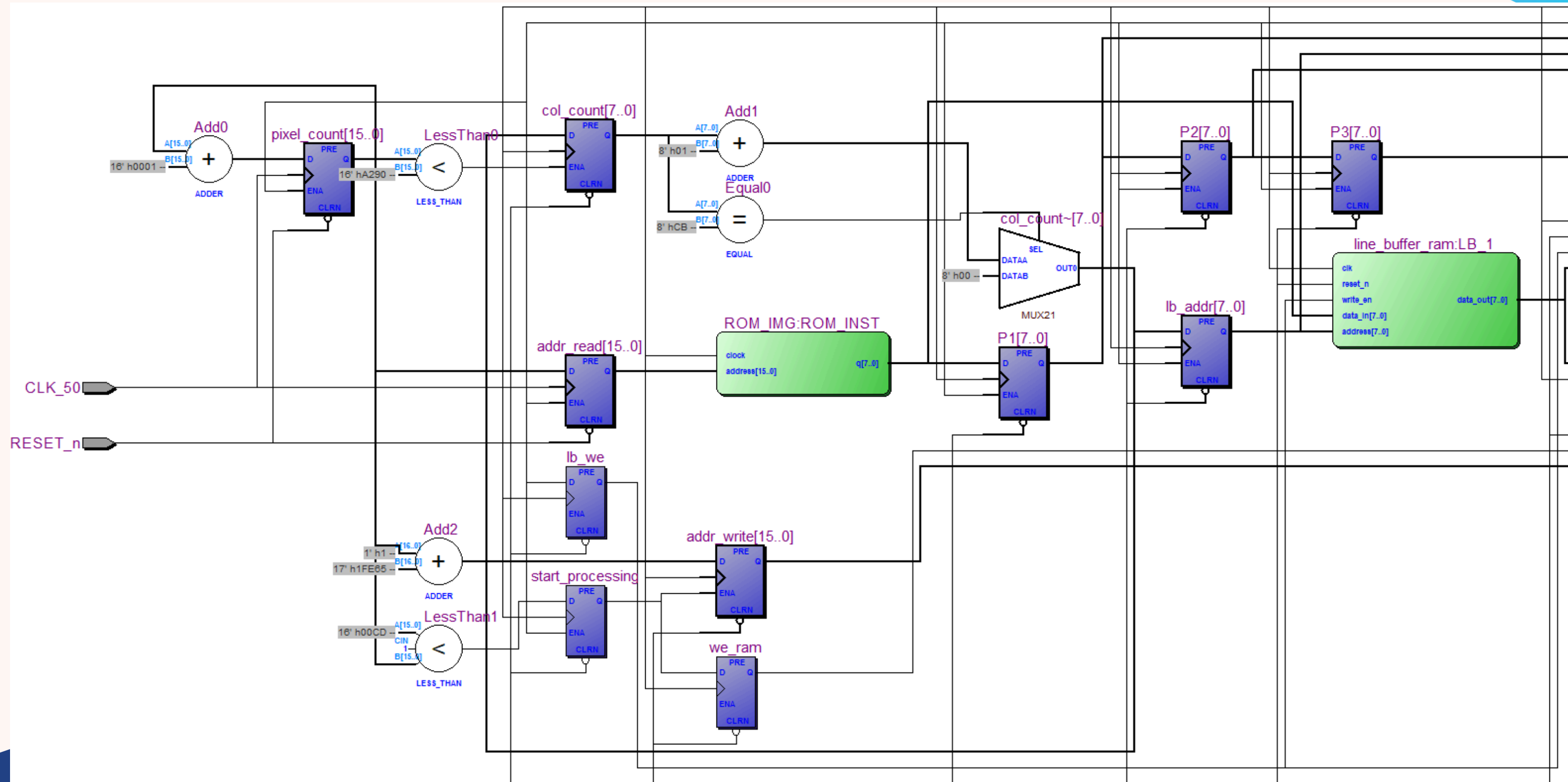
$\text{GENERATE_MIF}(\text{INPUT_IMAGE_FILE}, \text{OUTPUT_MIF_FILE}, \text{BITS_PER_PIXEL})$



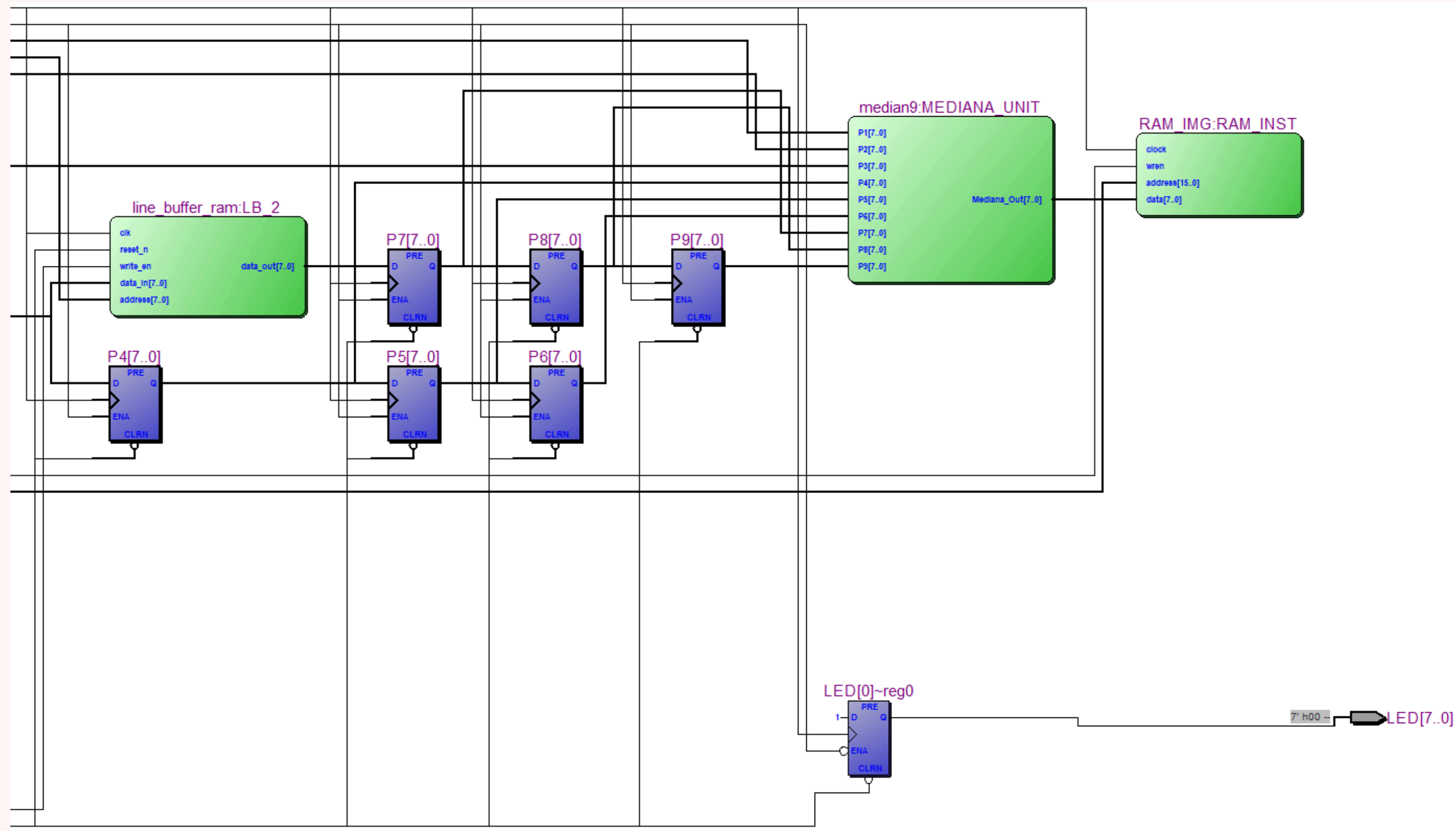
ARQUITECTURA



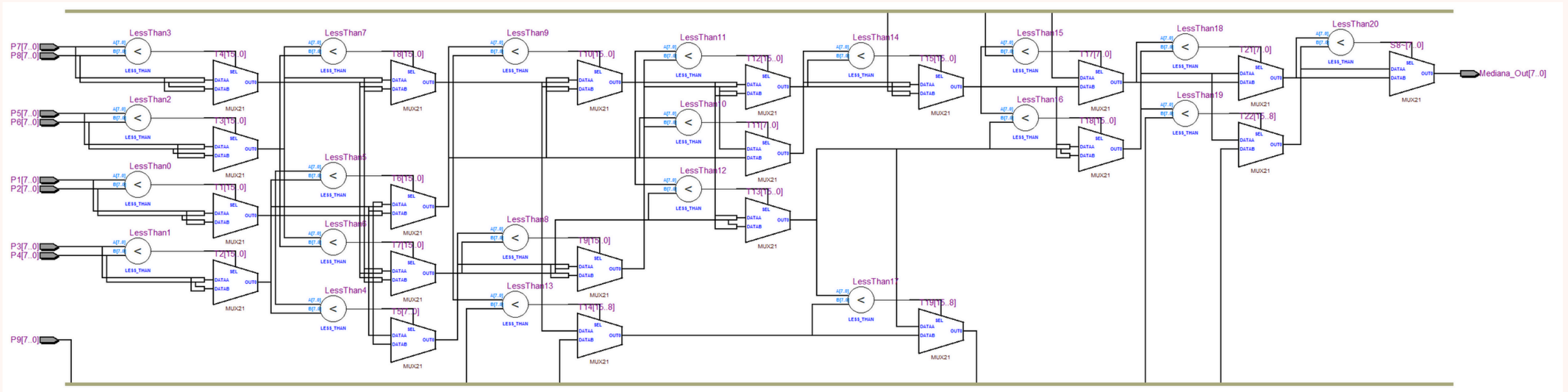
ARQUITECTURA



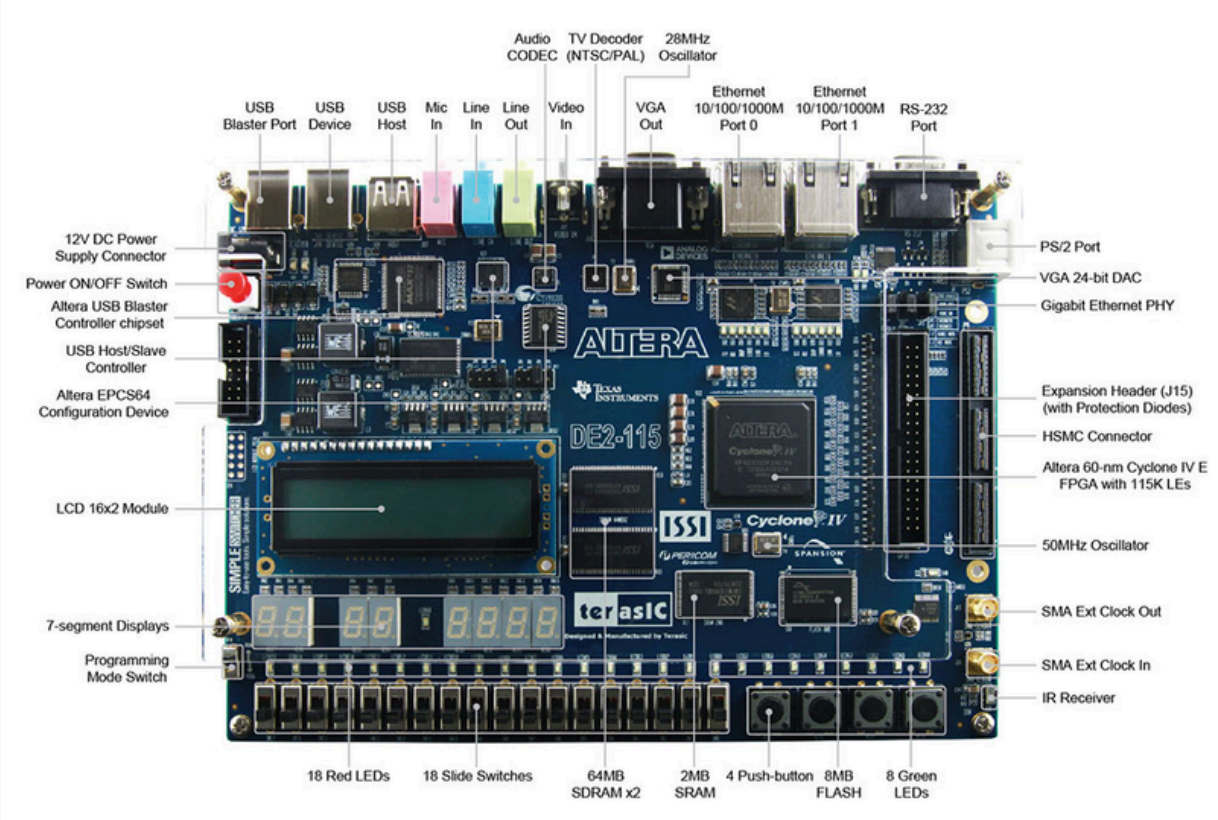
ARQUITECTURA



ARQUITECTURA MEDIANA



PROCESAMIENTO DE IMAGEN SALIDA

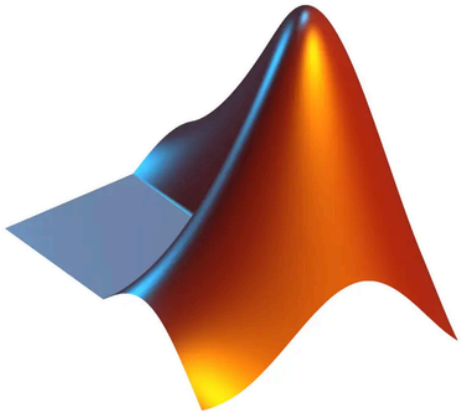


GENERATE_PNG_FROM_MIF

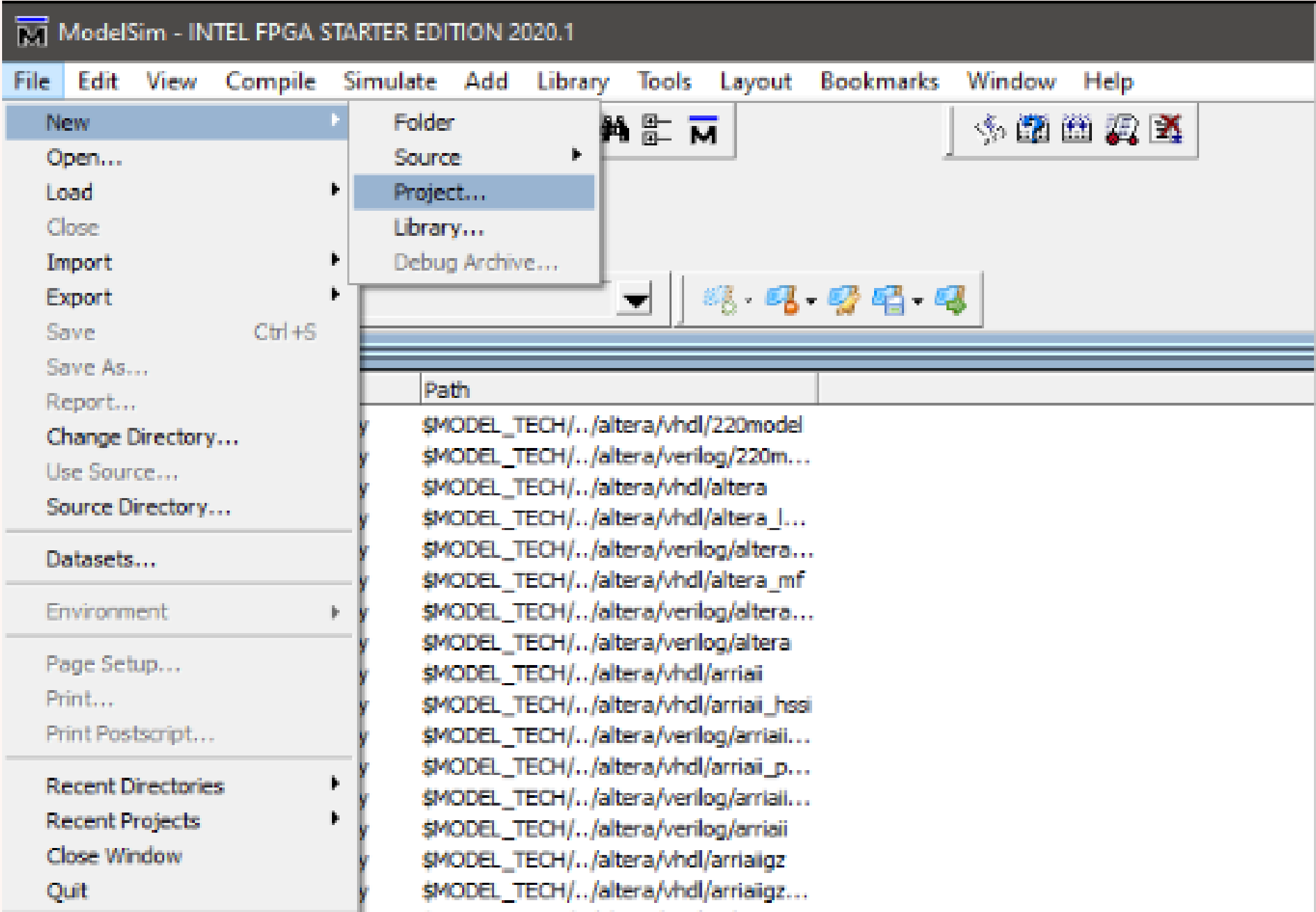
GENERATE_PNG_FROM_MIF (INPUT_MIF_FILENAME, OUTPUT_IMAGE_FILENAME, FILAS, COLUMNAS)

ARCHIVO MIF

MATLAB



SIMULACION EN MODELSIM

- 

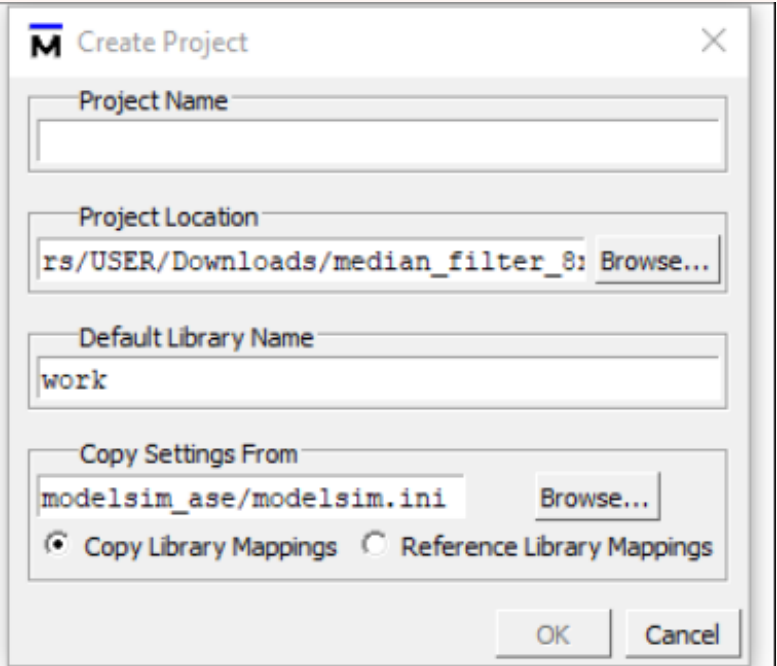
ModelSim - INTEL FPGA STARTER EDITION 2020.1

File Edit View Compile Simulate Add Library Tools Layout Bookmarks Window Help

New Open... Load Close Import Export Save Save As... Report... Change Directory... Use Source... Source Directory... Datasets... Environment Page Setup... Print... Print Postscript... Recent Directories Recent Projects Close Window Quit

Folder Source Project... Library... Debug Archive...

Path

 - \$MODEL_TECH/..altera/vhdl/220model
 - \$MODEL_TECH/..altera/verilog/220m...
 - \$MODEL_TECH/..altera/vhdl/altera
 - \$MODEL_TECH/..altera/vhdl/altera_...
 - \$MODEL_TECH/..altera/verilog/altera...
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- 

Create Project

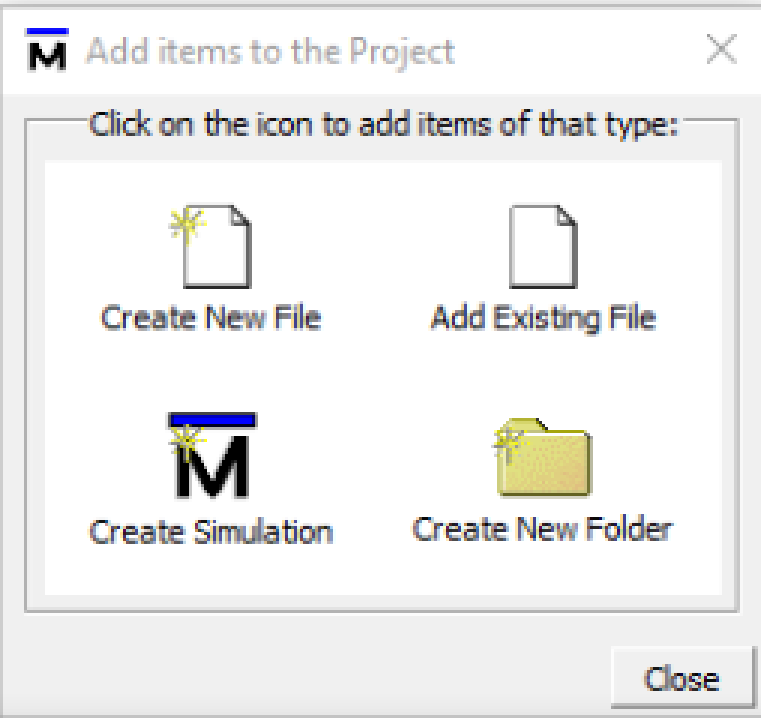
Project Name

Project Location
rs/USER/Downloads/median_filter_8: Browse...

Default Library Name
work

Copy Settings From
modelsim_ase/modelsim.ini Browse...

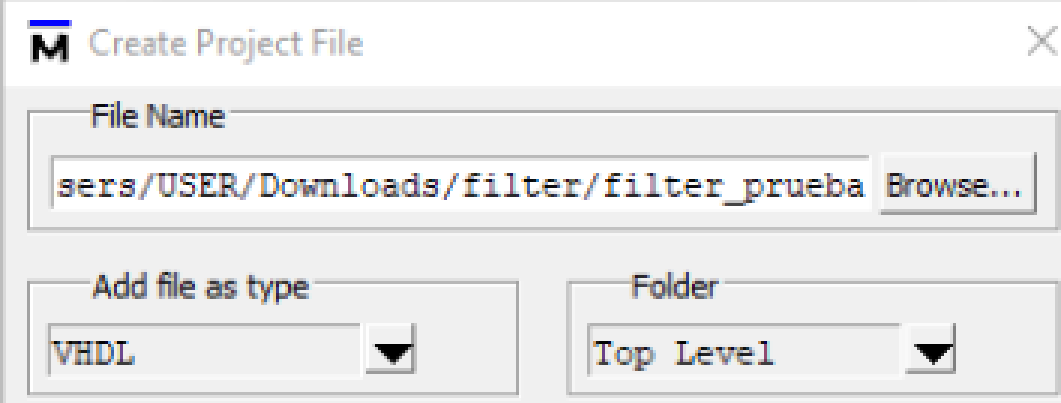
☒ Copy Library Mappings ☐ Reference Library Mappings

OK Cancel
- 

Add items to the Project

Click on the icon to add items of that type:

 - Create New File
 - Add Existing File
 - Create Simulation
 - Create New Folder

Close
- 

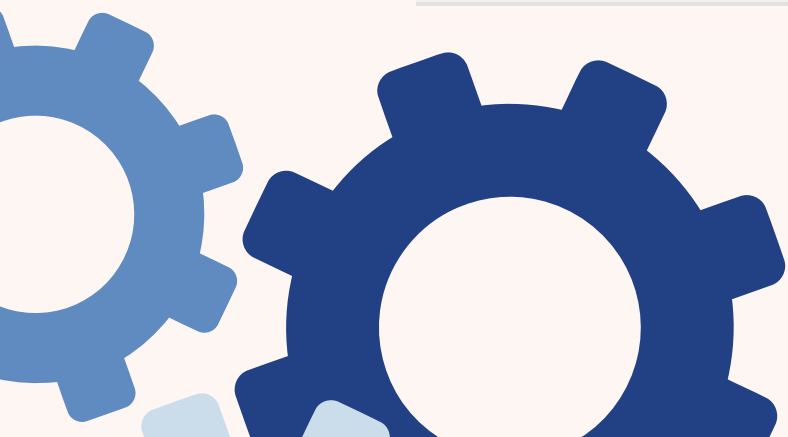
Create Project File

File Name
sers/USER/Downloads/filter/filter_prueba Browse...

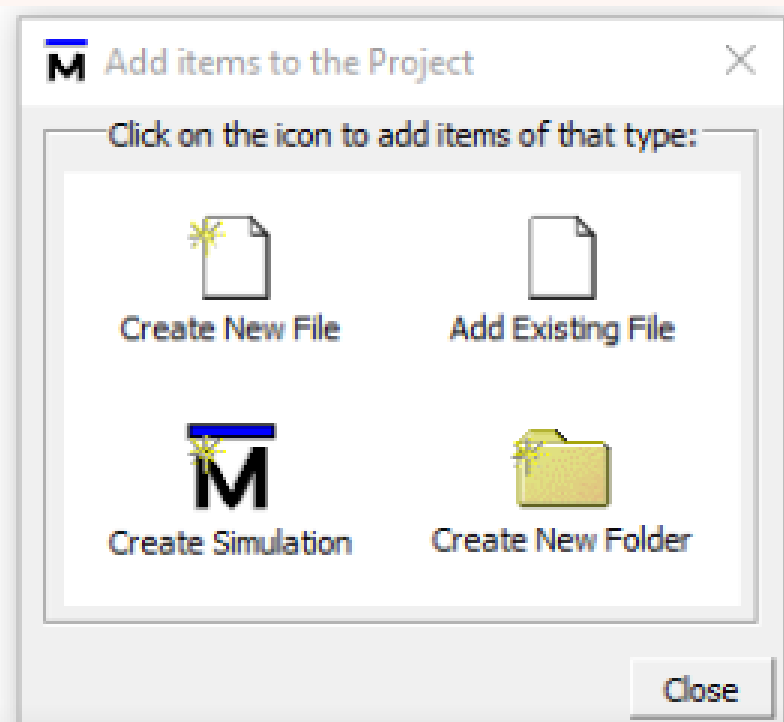
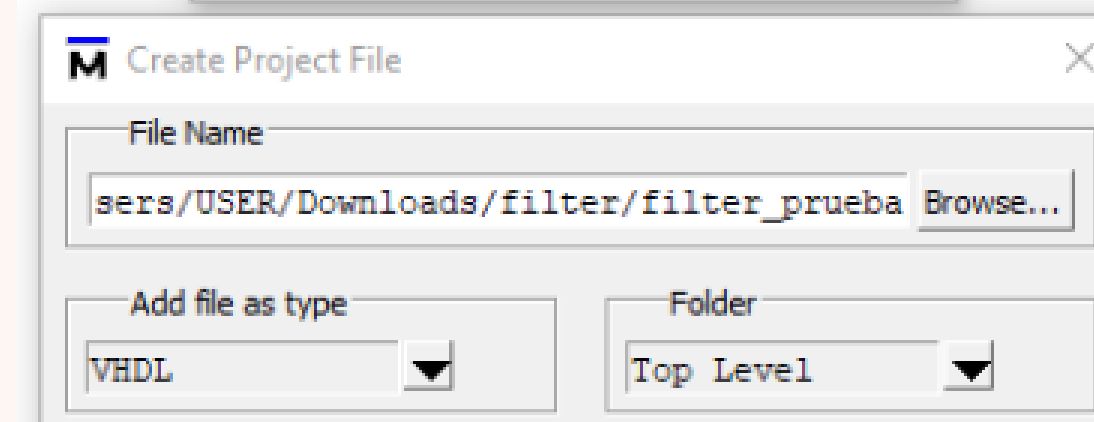
Add file as type
VHDL

Folder
Top Level

1

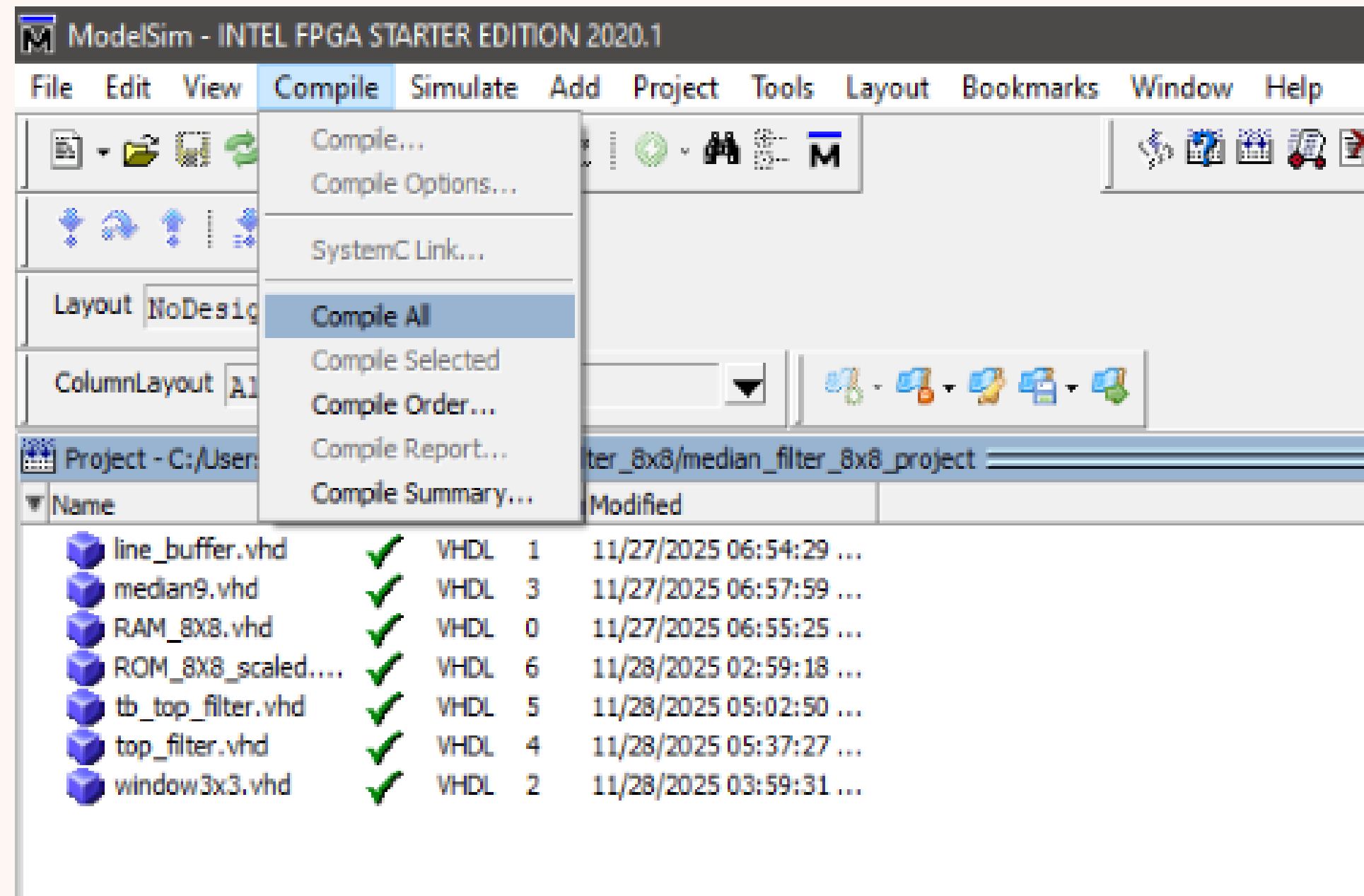


3

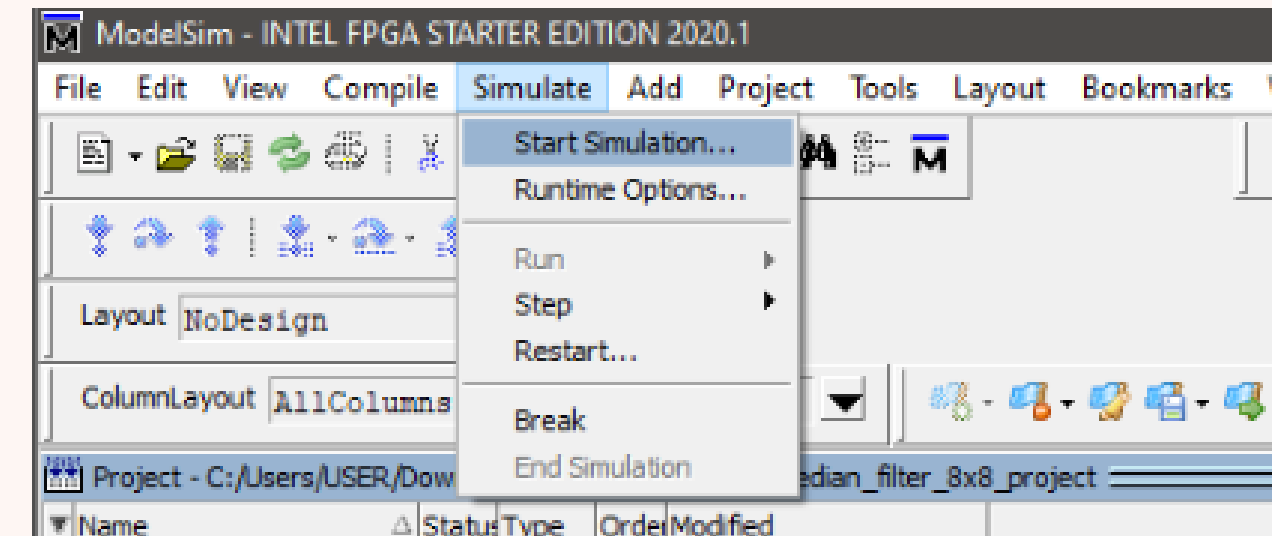


SIMULACION EN MODELSIM

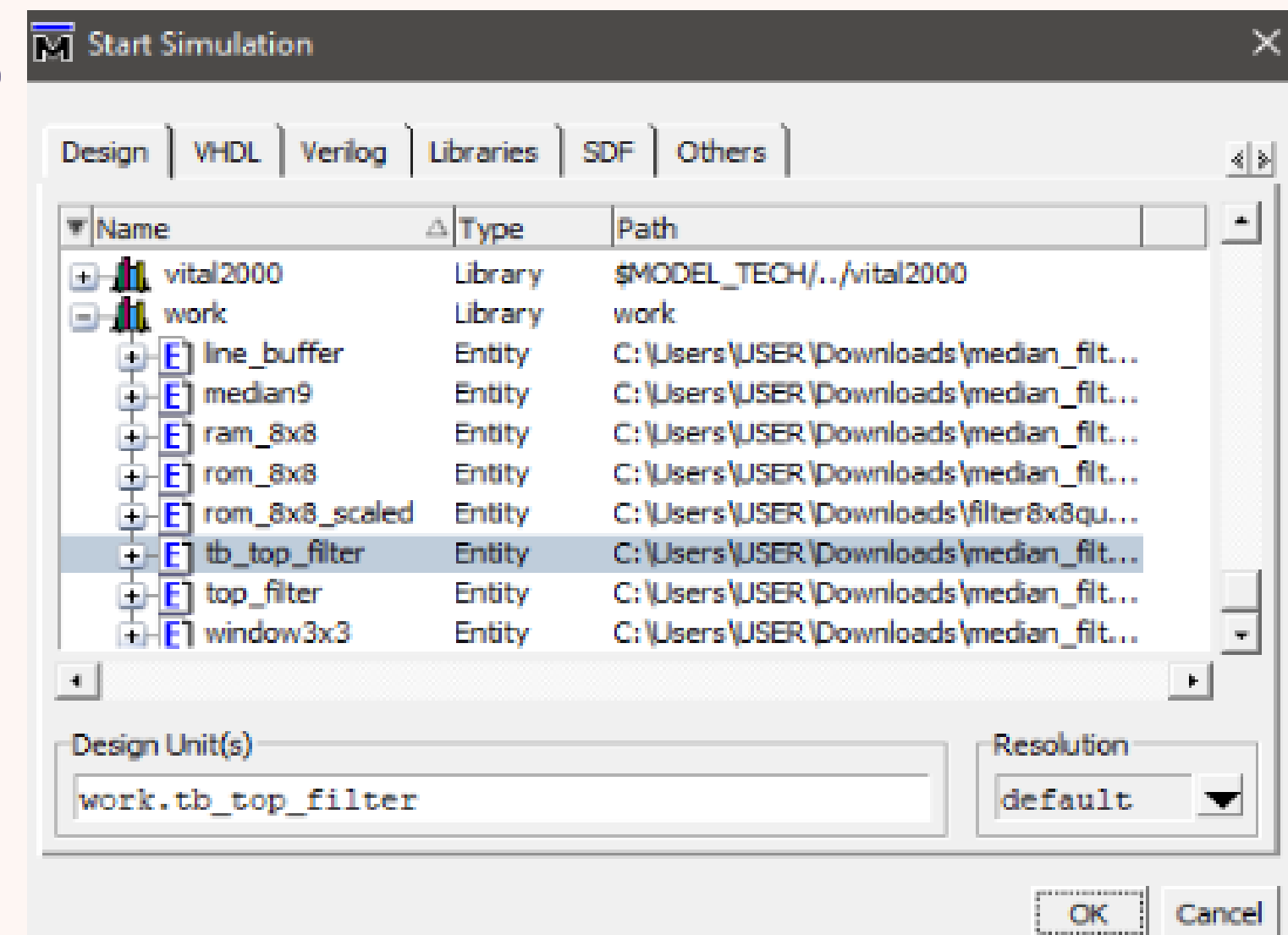
4



5



6



SIMULACION EN MODELSIM

7

```
Transcript
couldn't open "transcript": permission denied
# Reading pref.tcl
# Loading project filter_prueba
# reading C:/intelFPGA_lite/20.1/modelsim_ase/win32aloem/./modelsim.ini
# Loading project median_filter_8x8_project
ModelSim> vsim -gui work.tb_top_filter
# vsim -gui work.tb_top_filter
# Start time: 10:48:05 on Nov 29, 2025
# Loading std.standard
# Loading std.textio(body)
# Loading ieee.std_logic_1164(body)
# Loading ieee.numeric_std(body)
# Loading work.tb_top_filter(sim)
# Loading work.rom_8x8_scaled(rtl)
# Loading work.top_filter(rtl)
# Loading work.median9(rtl)
VSIM 2> add wave *
VSIM 3> add wave sim:/tb_top_filter/DUT/*
```

IMAGEN DE MATRIZ 8X8

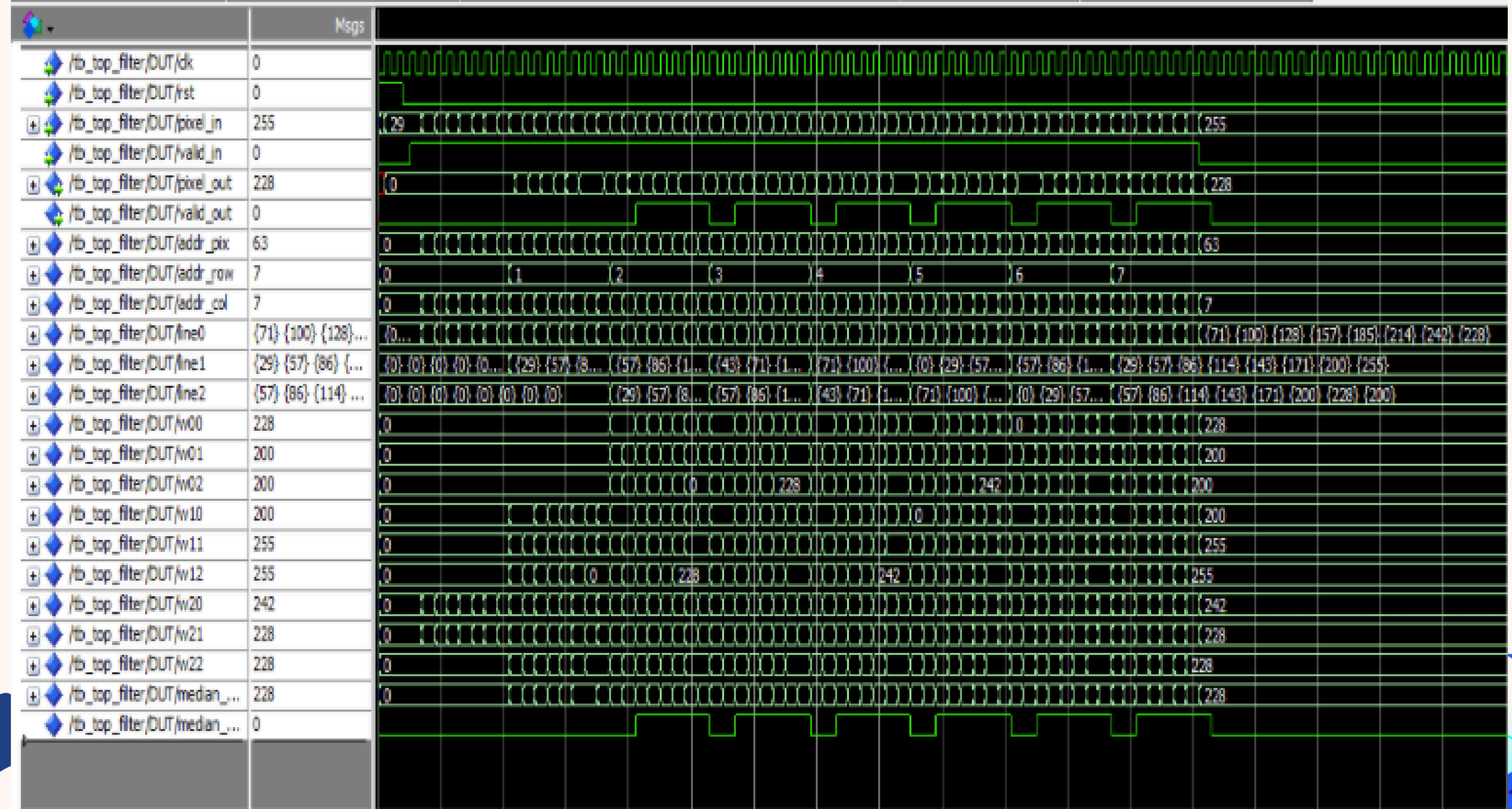
29	57	86	114	143	171	200	228
57	86	114	143	171	200	228	255
43	71	100	128	157	185	214	242
71	100	128	157	185	214	242	255
0	29	57	86	114	143	171	200
57	86	114	143	171	200	228	255
29	57	86	114	143	171	200	228
71	100	128	157	185	214	242	255



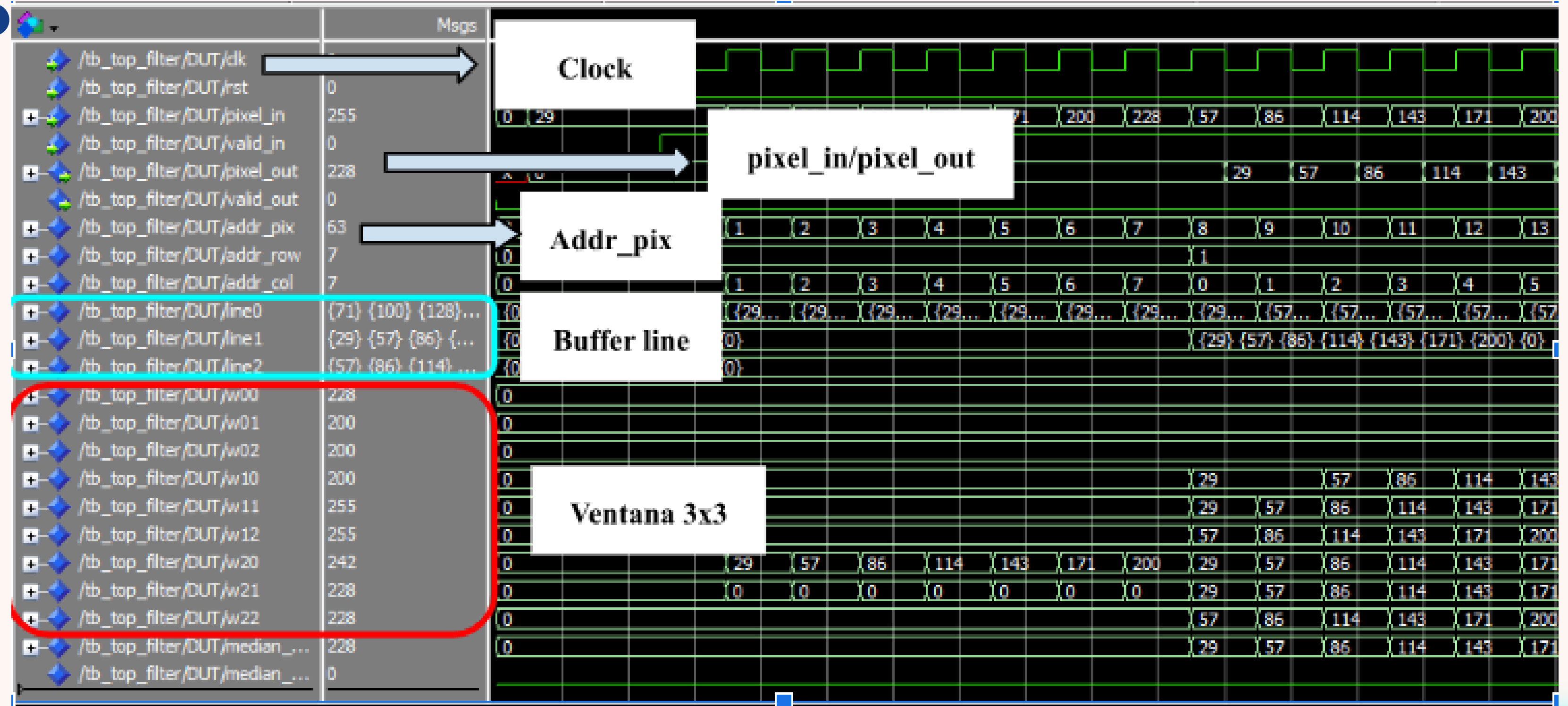
Matriz Original Escalada a Grises 0-255							
29	57	86	114	143	171	200	228
57	86	114	143	171	200	228	255
43	71	100	128	157	185	214	242
71	100	128	157	185	214	242	255
0	29	57	86	114	143	171	200
57	86	114	143	171	200	228	255
29	57	86	114	143	171	200	228
71	100	128	157	185	214	242	255

Filtro Mediana 3x3 con 2 line buffers (Valid-Only)							
29	57	86	114	143	171	200	228
57	86	114	143	171	200	228	255
43	71	100	128	157	185	214	242
71	86	114	143	171	200	228	255
0	71	100	128	157	185	214	200
57	71	100	128	157	185	214	255
29	57	86	114	143	171	200	228
71	86	114	143	171	200	228	255

SIMULACION EN MODELSIM



SIMULACION EN MODELSIM



SIMULACION EN MODELSIM

PIXEL 57 UBICADO EN LA POSICIÓN 34.

	Msgs	
/tb_top_filter/DUT/dk	1	
/tb_top_filter/DUT/rst	0	
+ /tb_top_filter/DUT/pixel_in	57	0 29 57 86
/tb_top_filter/DUT/valid_in	1	
+ /tb_top_filter/DUT/pixel_out	100	71 100 128
/tb_top_filter/DUT/valid_out	1	
+ /tb_top_filter/DUT/addr_pix	34	32 33 34 35
+ /tb_top_filter/DUT/addr_row	4	4
+ /tb_top_filter/DUT/addr_col	2	0 1 2 3
+ /tb_top_filter/DUT/line0	{0} {29} {128} {...	{71} {100} {128} {157} {185} {214} {242} {242}
+ /tb_top_filter/DUT/line1	{71} {100} {128} {...	{43} {71} {100} {128} {157} {185} {214} {255}
+ /tb_top_filter/DUT/line2	{43} {71} {100} {...	
+ /tb_top_filter/DUT/w00	71	43 71 100
+ /tb_top_filter/DUT/w01	100	43 71 100 128
+ /tb_top_filter/DUT/w02	128	71 100 128 157
+ /tb_top_filter/DUT/w10	100	71 100 128
+ /tb_top_filter/DUT/w11	128	71 100 128 157
+ /tb_top_filter/DUT/w12	157	100 128 157 185
+ /tb_top_filter/DUT/w20	29	71 0 29 57
+ /tb_top_filter/DUT/w21	128	71 100 128 157
+ /tb_top_filter/DUT/w22	157	100 128 157 185
+ /tb_top_filter/DUT/median_...	128	71 100 128 157
+ /tb_top_filter/DUT/median_...	1	

SIMULACION EN MODELSIM

PIXEL 57 UBICADO EN LA POSICIÓN 34.

	Msgs	
/tb_top_filter/DUT/dk	1	
/tb_top_filter/DUT/rst	0	
+ /tb_top_filter/DUT/pixel_in	57	0 29 57 86
/tb_top_filter/DUT/valid_in	1	
+ /tb_top_filter/DUT/pixel_out	100	71 100 128
/tb_top_filter/DUT/valid_out	1	
+ /tb_top_filter/DUT/addr_pix	34	32 33 34 35
+ /tb_top_filter/DUT/addr_row	4	4
+ /tb_top_filter/DUT/addr_col	2	0 1 2 3
+ /tb_top_filter/DUT/line0	{0} {29} {128} {...	{71} {100} {128} {157} {185} {214} {242} {242}
+ /tb_top_filter/DUT/line1	{71} {100} {128} {...	{43} {71} {100} {128} {157} {185} {214} {255}
+ /tb_top_filter/DUT/line2	{43} {71} {100} {...	43 71 100 128 157 185
+ /tb_top_filter/DUT/w00	71	43 71 100 128 157 185
+ /tb_top_filter/DUT/w01	100	71 100 128 157 185
+ /tb_top_filter/DUT/w02	128	71 100 128 157 185
+ /tb_top_filter/DUT/w10	100	71 100 128 157 185
+ /tb_top_filter/DUT/w11	128	71 100 128 157 185
+ /tb_top_filter/DUT/w12	157	71 100 128 157 185
+ /tb_top_filter/DUT/w20	29	71 100 128 157 185
+ /tb_top_filter/DUT/w21	128	71 100 128 157 185
+ /tb_top_filter/DUT/w22	157	71 100 128 157 185
+ /tb_top_filter/DUT/median_...	128	71 100 128 157 185
+ /tb_top_filter/DUT/median_...	1	

SIMULACION EN MODELSIM

PIXEL 57 UBICADO EN LA POSICIÓN 34.

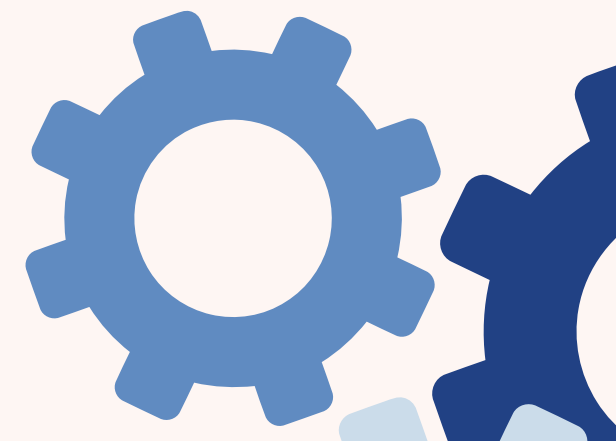
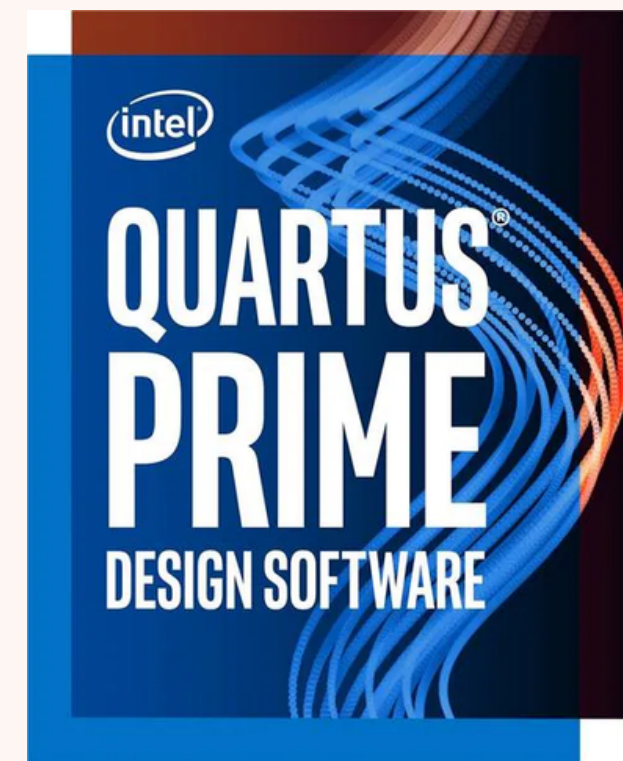
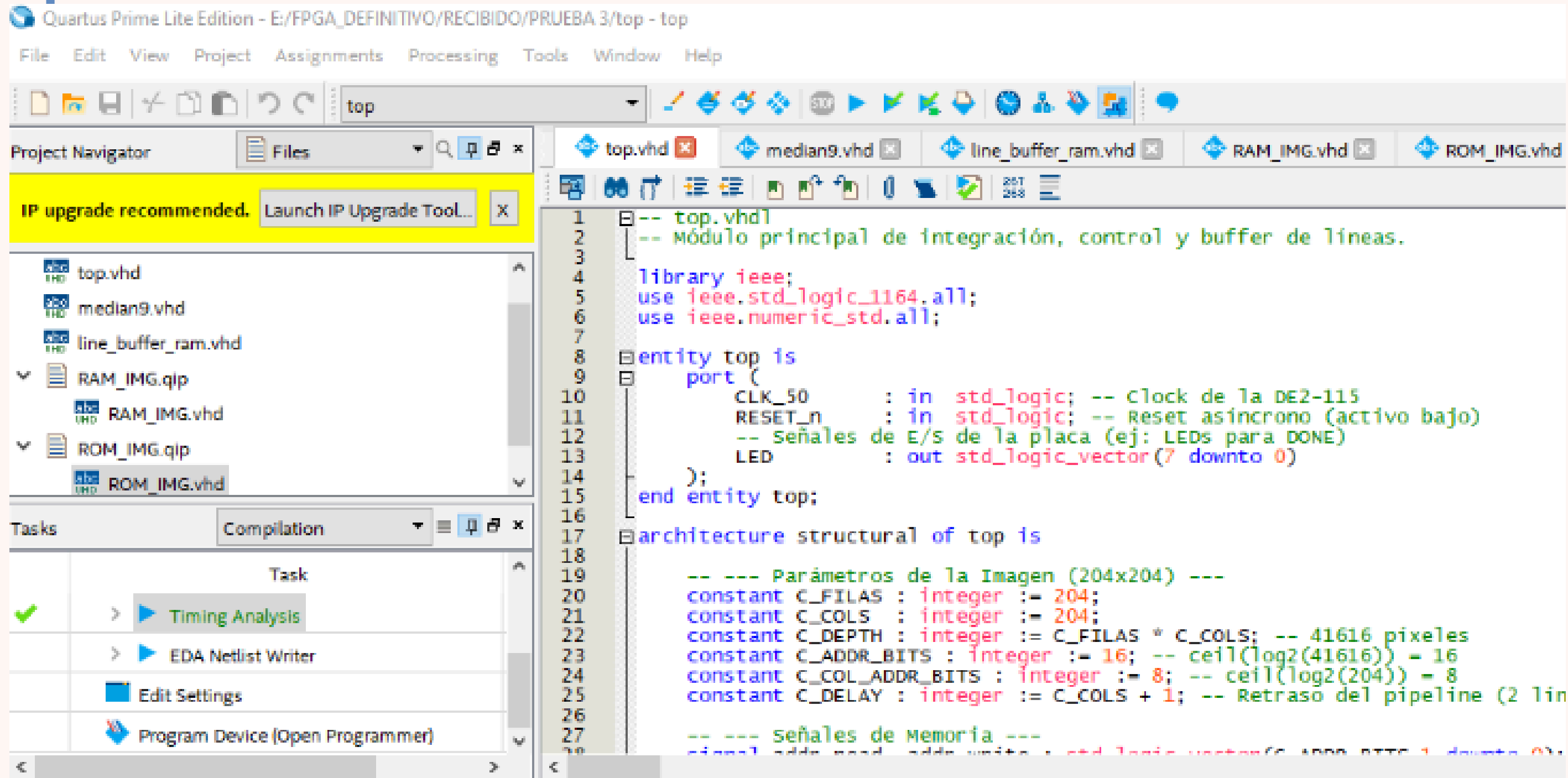
	Msgs	
/tb_top_filter/DUT/dk	1	
/tb_top_filter/DUT/rst	0	
+ /tb_top_filter/DUT/pixel_in	57	0 29 57 86
/tb_top_filter/DUT/valid_in	1	
+ /tb_top_filter/DUT/pixel_out	100	71 100 128
/tb_top_filter/DUT/valid_out	1	
+ /tb_top_filter/DUT/addr_pix	34	32 33 34 35
+ /tb_top_filter/DUT/addr_row	4	4
+ /tb_top_filter/DUT/addr_col	2	0 1 2 3
+ /tb_top_filter/DUT/line0	{0} {29} {128} {...	{71} {100} {128} {157} {185} {214} {242} {242}
+ /tb_top_filter/DUT/line1	{71} {100} {128} {...	{43} {71} {100} {128} {157} {185} {214} {255}
+ /tb_top_filter/DUT/line2	{43} {71} {100} {...	43 71 100 128 157 185
+ /tb_top_filter/DUT/w00	71	43 71 100 128 157 185
+ /tb_top_filter/DUT/w01	100	71 100 128 157 185
+ /tb_top_filter/DUT/w02	128	71 100 128 157 185
+ /tb_top_filter/DUT/w10	100	71 100 128 157 185
+ /tb_top_filter/DUT/w11	128	71 100 128 157 185
+ /tb_top_filter/DUT/w12	157	71 100 128 157 185
+ /tb_top_filter/DUT/w20	29	71 100 128 157 185
+ /tb_top_filter/DUT/w21	128	71 100 128 157 185
+ /tb_top_filter/DUT/w22	157	71 100 128 157 185
+ /tb_top_filter/DUT/median_...	128	71 100 128 157 185
+ /tb_top_filter/DUT/median_...	1	

SIMULACION EN MODELSIM

PIXEL 128 UBICADO EN LA POSICIÓN 19.

	Mags	
 /tb_top_filter/DUT/clk	1	
 /tb_top_filter/DUT/rst	0	
  /tb_top_filter/DUT/pixel_in	128	228 255 43 71 100 128 157 185 214 242
 /tb_top_filter/DUT/valid_in	1	
  /tb_top_filter/DUT/pixel_out	114	171 0 57 86 114 143 171 200
 /tb_top_filter/DUT/valid_out	1	
  /tb_top_filter/DUT/addr_pix	19	14 15 16 17 18 19 20 21 22 23
 /tb_top_filter/DUT/addr_row	2	1 2
 /tb_top_filter/DUT/addr_col	3	6 7 0 1 2 3 4 5 6 7
  /tb_top_filter/DUT/line0	{43} {71} {100} ...	{57} ... {57} {86} { ... {57} {86} { ... {43} {86} { ... {43} {71} { ... {43} {71} {1... {43} {71} { ... {43} {71} { ... {43}
  /tb_top_filter/DUT/line1	{57} {86} {114} ...	{29} {57} {86} {114} ... {57} {86} {114} {143} {171} {200} {228} {228}
  /tb_top_filter/DUT/line2	{29} {57} {86} { ...	{0} {0} {0} {0} {0} {0... {29} {57} {86} {114} {143} {171} {200} {0}
 /tb_top_filter/DUT/w00	86	0 29 57 86 114 143 171 200
 /tb_top_filter/DUT/w01	114	0 29 57 86 114 143 171 200 0
 /tb_top_filter/DUT/w02	143	0 57 86 114 143 171 200 0
 /tb_top_filter/DUT/w10	114	171 200 57 86 114 143 171 200 228
 /tb_top_filter/DUT/w11	143	200 0 57 86 114 143 171 200 228
 /tb_top_filter/DUT/w12	171	0 86 114 143 171 200 228
 /tb_top_filter/DUT/w20	100	200 228 57 43 71 100 128 157 185 214
 /tb_top_filter/DUT/w21	143	200 228 57 86 114 143 171 200 228 255
 /tb_top_filter/DUT/w22	171	228 86 114 143 171 200 228 255
  /tb_top_filter/DUT/median_...	143	171 0 57 86 114 143 171 200 200 228
 /tb_top_filter/DUT/median_...	1	

IMPLEMENTACION



ASIGNACIÓN DE PINES

USO DEL MANUAL



```
entity top is
  port (
    CLK_50    : in std_logic;      -- Clock físico de la DE2-115
    RESET_n   : in std_logic;      -- Reset por pulsador o switch
    LED       : out std_logic_vector(7 downto 0) -- LEDs de la placa
  );
end entity top;
```

Table 4-5 Pin Assignments for Clock Inputs

Signal Name	FPGA Pin No.	Description	I/O Standard
CLOCK_50	PIN_Y2	50 MHz clock input	3.3V
CLOCK2_50	PIN_AG14	50 MHz clock input	3.3V
CLOCK3_50	PIN_AG15	50 MHz clock input	Depending on JP6
SMA_CLKOUT	PIN_AE23	External (SMA) clock output	Depending on JP6
SMA_CLKIN	PIN_AH14	External (SMA) clock input	3.3V

Table 4-2 Pin Assignments for Push-buttons

Signal Name	FPGA Pin No.	Description	I/O Standard
KEY[0]	PIN_M23	Push-button[0]	Depending on JP7
KEY[1]	PIN_M21	Push-button[1]	Depending on JP7
KEY[2]	PIN_N21	Push-button[2]	Depending on JP7
KEY[3]	PIN_R24	Push-button[3]	Depending on JP7

Table 4-3 Pin Assignments for LEDs

Signal Name	FPGA Pin No.	Description	I/O Standard
LEDR[0]	PIN_G19	LED Red[0]	2.5V
LEDR[1]	PIN_F19	LED Red[1]	2.5V
LEDR[2]	PIN_E19	LED Red[2]	2.5V
LEDR[3]	PIN_F21	LED Red[3]	2.5V
LEDR[4]	PIN_F18	LED Red[4]	2.5V
LEDR[5]	PIN_E18	LED Red[5]	2.5V
LEDR[6]	PIN_J19	LED Red[6]	2.5V
LEDR[7]	PIN_H19	LED Red[7]	2.5V
LEDR[8]	PIN_J17	LED Red[8]	2.5V
LEDR[9]	PIN_G17	LED Red[9]	2.5V
LEDR[10]	PIN_J15	LED Red[10]	2.5V
LEDR[11]	PIN_H16	LED Red[11]	2.5V
LEDR[12]	PIN_J16	LED Red[12]	2.5V
LEDR[13]	PIN_H17	LED Red[13]	2.5V
LEDR[14]	PIN_F15	LED Red[14]	2.5V

IMPLEMENTACION

sort

Report not available

Groups

Report

Files

Early Pin Planning

Early Pin Planning

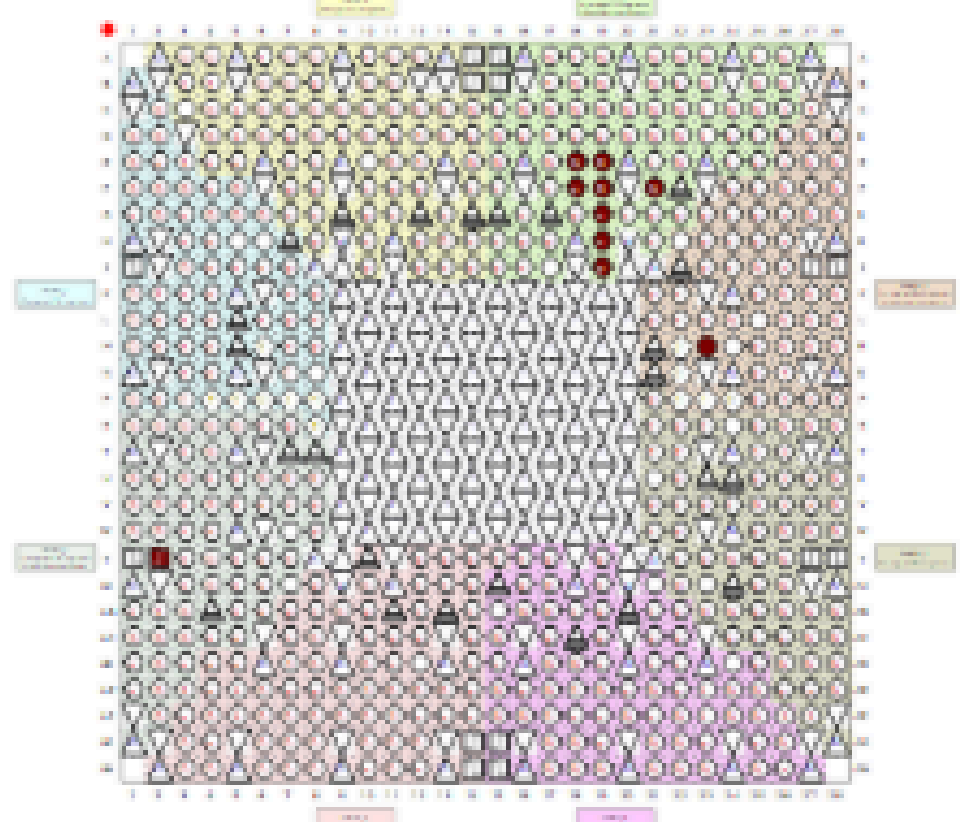
Run I/O Assignment

Export Pin Assignment

Pin Finder...

Top View - Wire Bond

Cyclone IV E - EP4CE115F29C7



Pin Legend

Symbol

U

U

F

U

R

Other configura...

DEV_OE

DEV_CLR

DIFF_n

DIFF_p

DQ

DQS

CLK_n

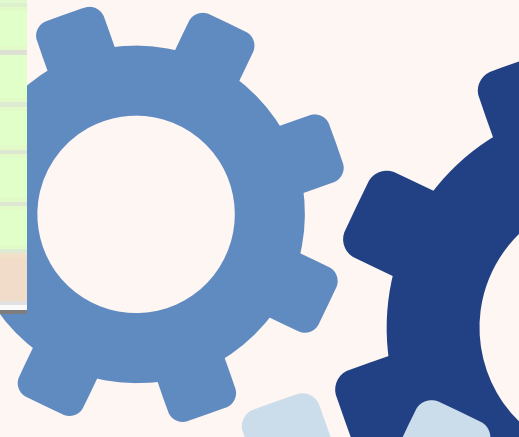
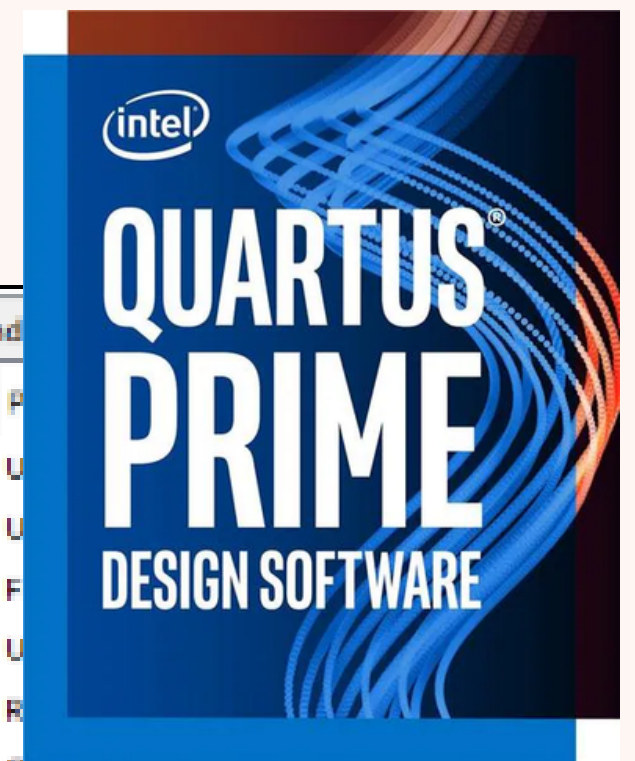
CLK_p

Named: *

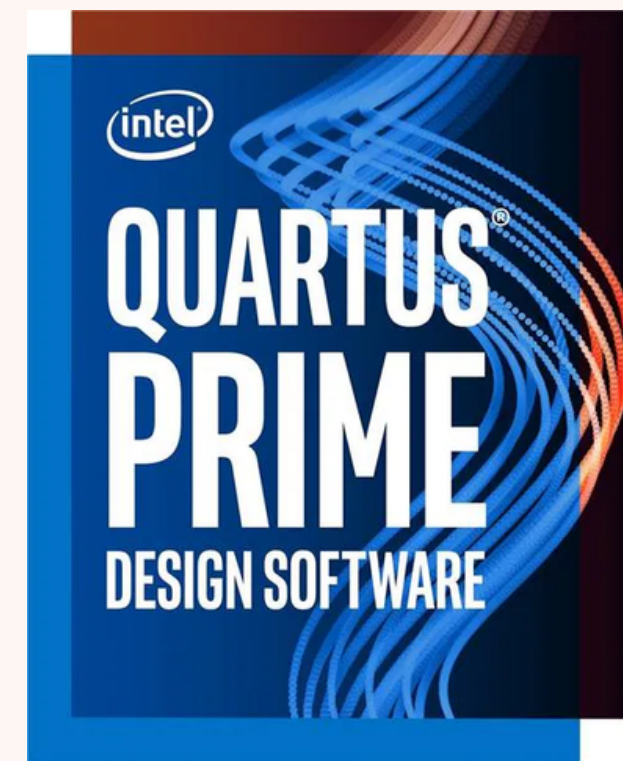
Edit

Filter: P

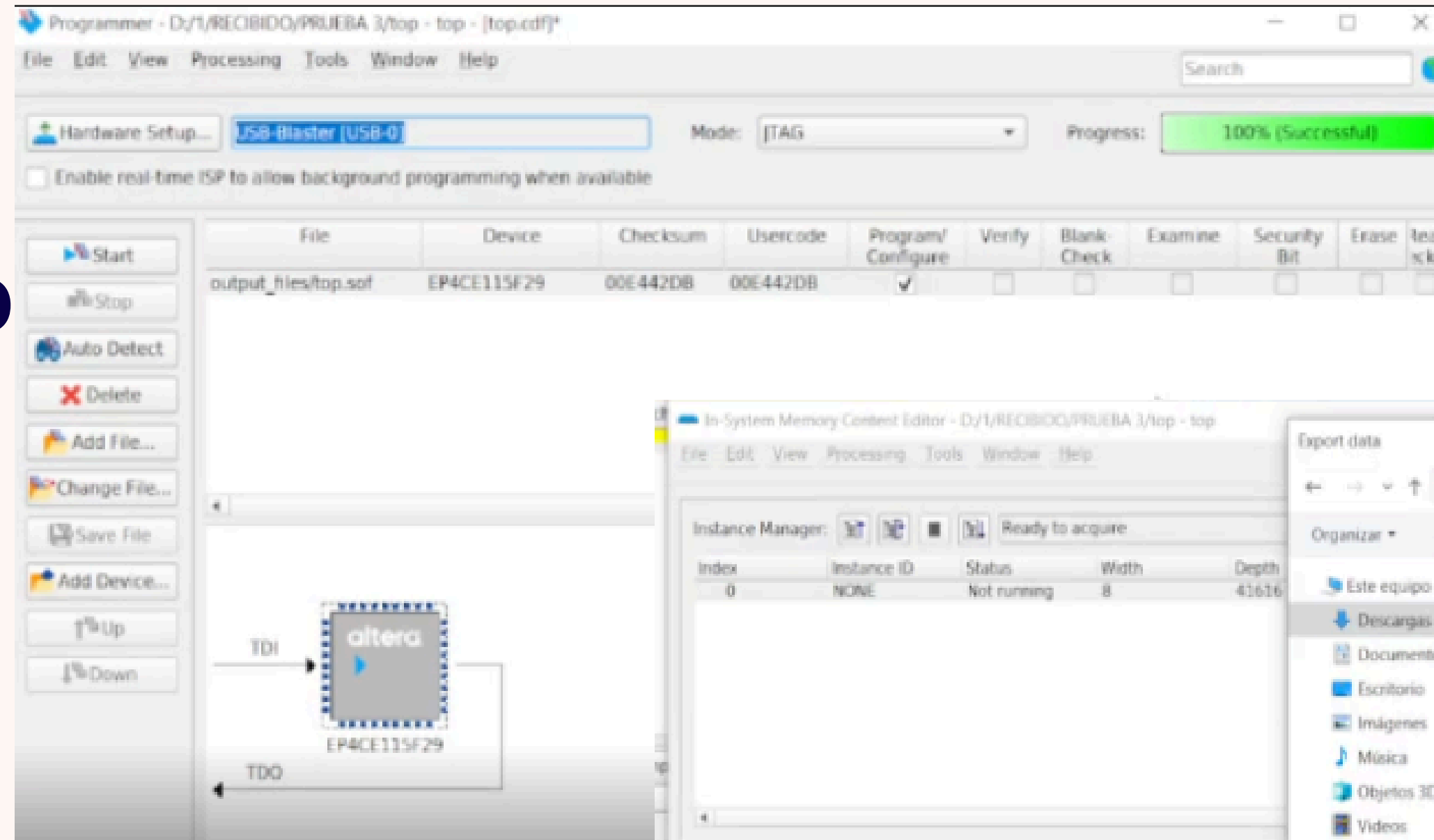
Node Name	Direction	Location	I/O Bank	VREF Group	Filter Location	I/O Standard	Reserved	Current Strength	Slew Rate	Differential Pair	Strict Pres
altera_reserved_tms	Input				PIN_P8	2.5 V (default)		8mA (default)			
CLK_50	Input	PIN_Y2	2	B2_N0	PIN_Y2	2.5 V		8mA (default)			
LED[7]	Output	PIN_H19	7	B7_N2	PIN_H19	2.5 V		8mA (default)	2 (default)		
LED[6]	Output	PIN_J19	7	B7_N2	PIN_J19	2.5 V		8mA (default)	2 (default)		
LED[5]	Output	PIN_E18	7	B7_N1	PIN_E18	2.5 V		8mA (default)	2 (default)		
LED[4]	Output	PIN_F18	7	B7_N1	PIN_F18	2.5 V		8mA (default)	2 (default)		
LED[3]	Output	PIN_F21	7	B7_N0	PIN_F21	2.5 V		8mA (default)	2 (default)		
LED[2]	Output	PIN_E19	7	B7_N0	PIN_E19	2.5 V		8mA (default)	2 (default)		
LED[1]	Output	PIN_F19	7	B7_N0	PIN_F19	2.5 V		8mA (default)	2 (default)		
LED[0]	Output	PIN_G19	7	B7_N2	PIN_G19	2.5 V		8mA (default)	2 (default)		
RESET_n	Input	PIN_M23	6	B6_N2	PIN_M23	2.5 V		8mA (default)			



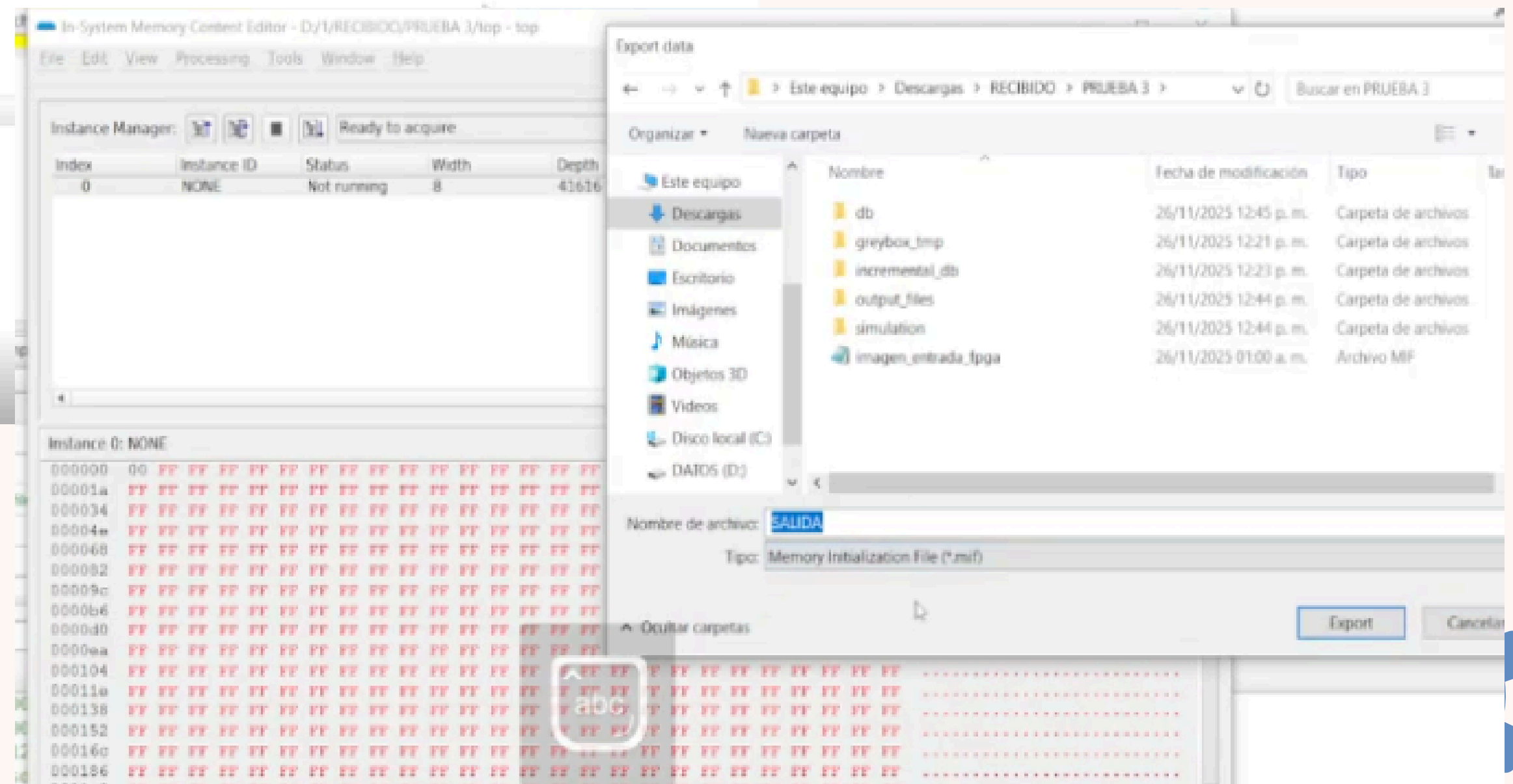
IMPLEMENTACION



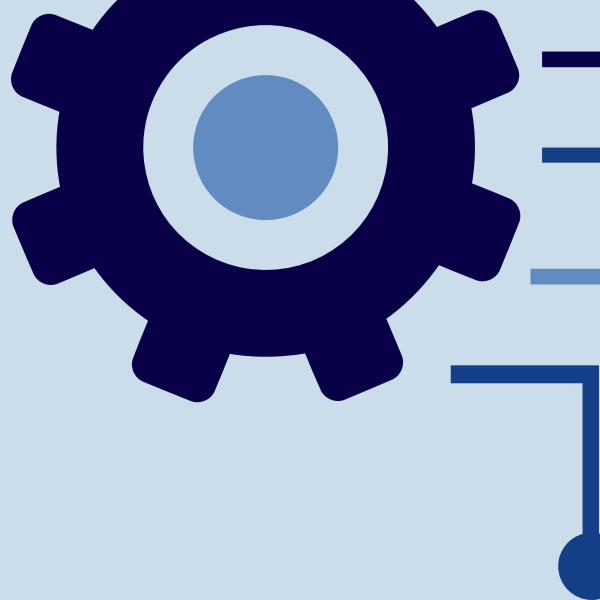
1ero



2do



RESULTADOS



SALIDA.mif: Bloc de notas

ArchivoEdiciónFormatoVerAyuda

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-- Quartus Prime generated Memory Initialization File (.mif)

WIDTH=8;
DEPTH=41616;

ADDRESS_RADIX=HEX;
DATA_RADIX=BIN;

CONTENT BEGIN
0000 : 00000000;
[0001..1000] : 11111111;
1001 : 11111110;
1002 : 11011111;
1003 : 10111100;
[1041..1044] : 11010001;
1045 : 11110010;
[1046..1048] : 11111111;
104C : 11110101;
1050 : 11100111;
[1068..10F0] : 11110101;
10F1 : 11111011;
[10F2..1100] : 11111111;
110A : 11001001;
110B : 01110111;
110C : 01101110;
[110D..1110] : 01101111;
1111 : 10000111;
1112 : 10111100;
[1113..1114] : 11010011;
1115 : 10100111;
[1116..1119] : 11001001;
111A : 11110000;
[111B..111C] : 11111111;
1117 : 11101011;
1118 : 11001001;
1119 : 11000010;
[11BA..11BC] : 11001001;
11BD : 11010011;

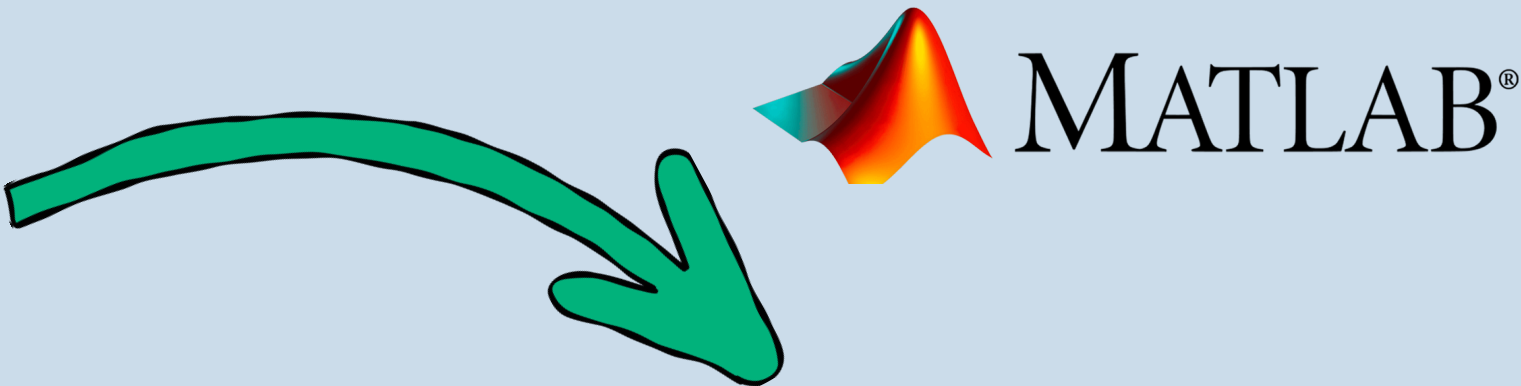


IMAGEN INICIAL CON
RUIDO



IMAGEN CON FILTRO
MEDIANA FPGA

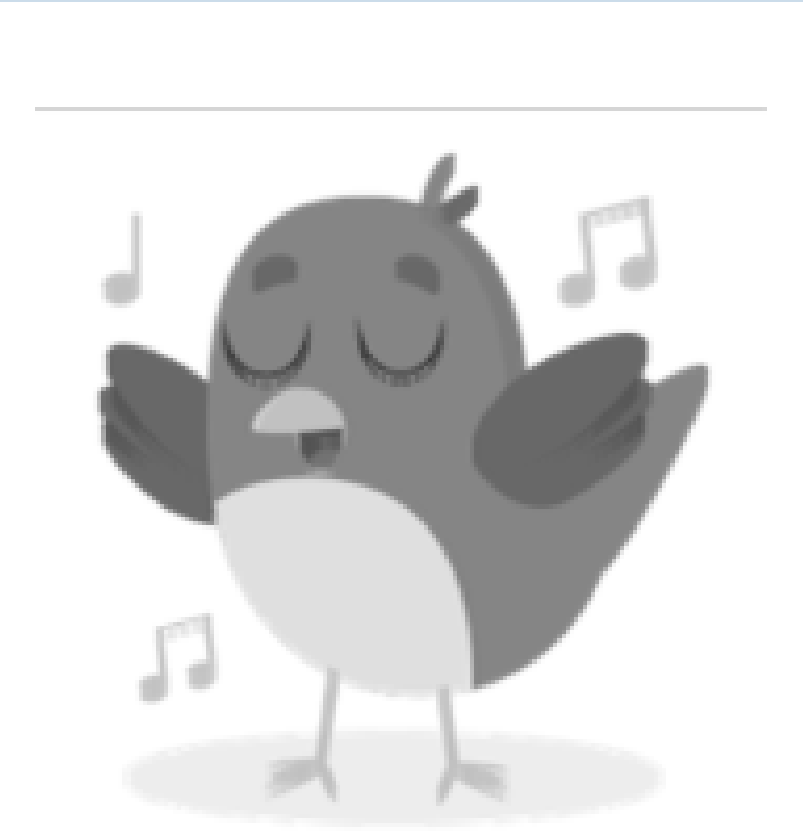
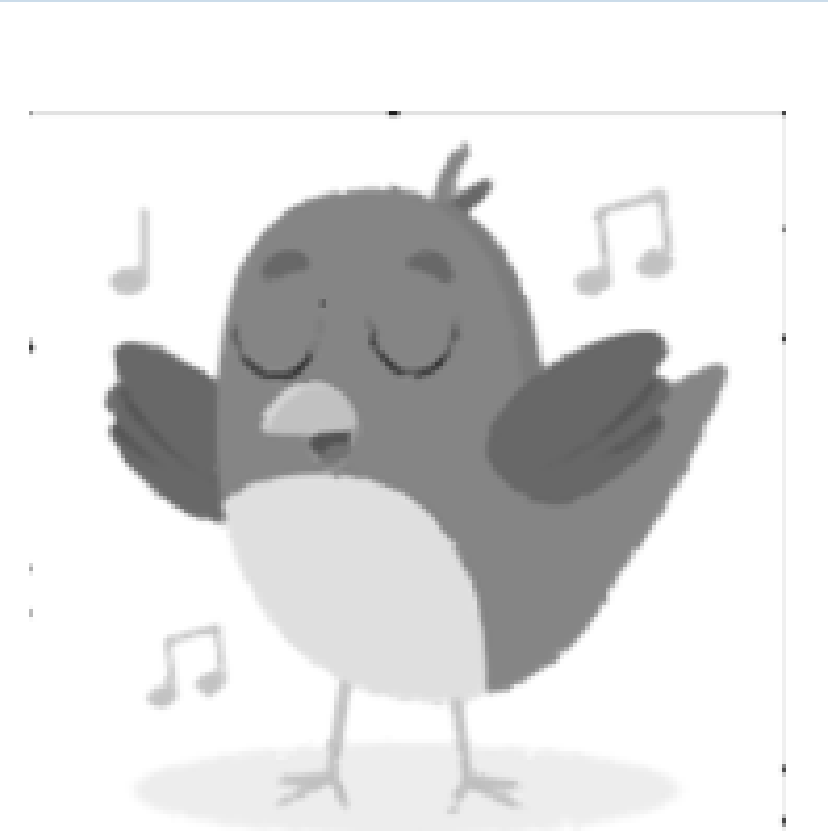
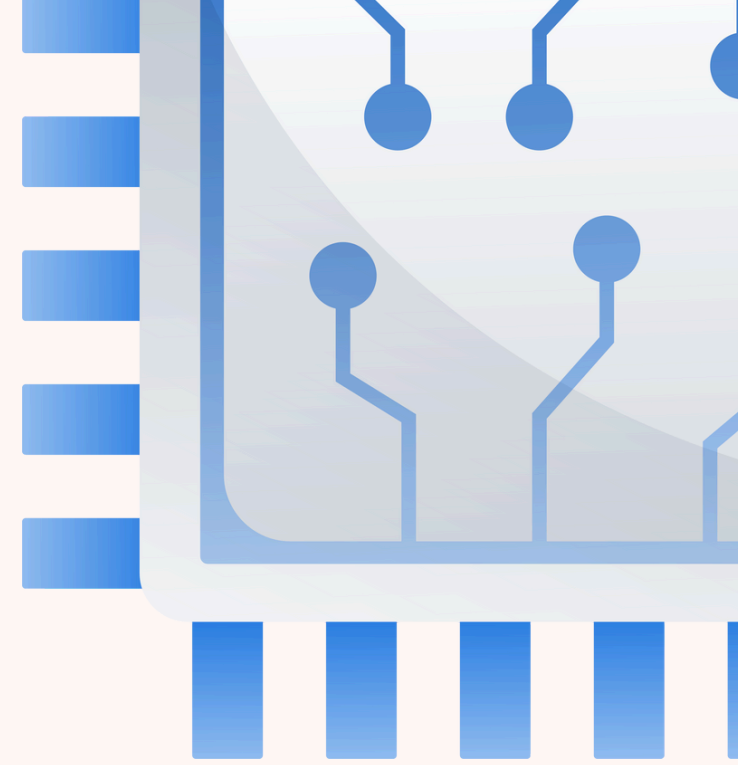
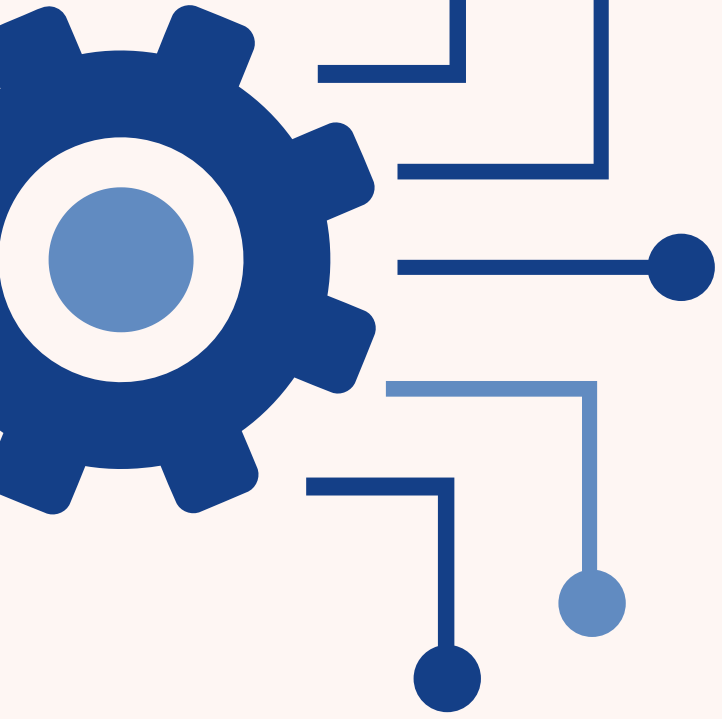


IMAGEN CON FILTRO
MEDIANA EN METLAB



CONCLUSION

- La implementación del filtro de mediana en la FPGA DE2-115 permitió procesar imágenes con ruido sal-y-pimienta de forma rápida gracias al diseño de nuestra arquitectura. Usar una imagen de 8×8 facilitó explicar la formación de la ventana 3×3 y el cálculo de la mediana mediante las formas de onda en ModelSim.
- MATLAB se empleó para generar archivos .MIF y verificar visualmente los resultados, validando el correcto funcionamiento del filtro tanto en simulación como en la FPGA.
- Durante la implementación se corrigieron errores derivados de la falta de módulos, especialmente los line buffers, esenciales para mantener un flujo de datos continuo y formar correctamente la ventana 3×3 .
- El diseño final es modular, escalable y puede adaptarse a imágenes de mayor resolución según la memoria disponible en la DE2-115.



GRACIAS

